

Yunhan Qiao

Ph.D. Candidate in Computer Science · Oregon State University

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RESEARCH PROFILE

Ph.D. candidate in Computer Science at Oregon State University, advised by Dr. Christopher Hundhausen. Research focuses on how generative AI assistants (e.g., GitHub Copilot) shape code comprehension across both computing education and professional software engineering, with a particular emphasis on brownfield programming and program understanding. Mixed-methods empirical researcher combining controlled experiments, qualitative interviews, and behavioral analysis to study GenAI adoption, comprehension trade-offs, and learning outcomes.

EDUCATION

Ph.D. in Computer Science, Oregon State University

Sep 2023 – Jun 2027 (expected)

Advisor: Dr. Christopher Hundhausen

*Research areas: GenAI-assisted code comprehension, computing education, empirical software engineering
Corvallis, Oregon*

M.S. in Software Engineering, Arizona State University

Aug 2021 – May 2023

Thesis: Networked System for Volumetric Athletic Coaching in Augmented Reality

Advisor: Dr. Robert LiKamWa

Tempe, Arizona

B.E. in Software Engineering, Guangxi University of Science and Technology

Sep 2016 – Jun 2020

Liuzhou, China

B.E. in Information Technology, Southern Cross University

Feb 2017 – Jun 2020

Gold Coast, Australia

PUBLICATIONS

Journal Articles

Yunhan Qiao, Md. Istiak Hossain Shihab, & Christopher Hundhausen (2026).

A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.

ACM Transactions on Computing Education (TOCE) — vol. 26, no. 2, pp. 1–33 [pdf]

Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, S. Chandrasekaran, Yunhan Qiao, & Summit Haque (2026).

Using Slackbot-Prompted Reflections to Promote Participation and Reduce Contribution Inequalities in Team Software Development Projects.

ACM Transactions on Computing Education (TOCE)

Conference Proceedings

Manish Motwani, Aakash Kulkarni, Yunhan Qiao, Matthew Davis, & Ziyan Chen (2025).

LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.

Proceedings of the IEEE International Conference on AI x Software Engineering (AIxSE) [pdf]

Md. Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, **Yunhan Qiao**, & Brian W. Mulanda (2025).
The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.
Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER), Vol. 1 — pp. 407–420 [pdf]

Under Submission / In Preparation

Yunhan Qiao, Christopher Hundhausen, & Md. Istiak Hossain Shihab (2027).
Teaching GenAI-Assisted Software Development in Computing Education: An Experimental Study.
Under submission, 58th ACM Technical Symposium on Computer Science Education (SIGCSE)

Yunhan Qiao, Md. Istiak Hossain Shihab, Ahsun Tariq, & Christopher Hundhausen (2027).
Unraveling the Codebase: Exploring GenAI Adoption for Program Comprehension Among Expert Developers in Brownfield Programming Tasks.
Under submission, 49th IEEE/ACM International Conference on Software Engineering (ICSE)

Yunhan Qiao, Md. Istiak Hossain Shihab, Summit Haque, & Christopher Hundhausen (2027).
Code Comprehension with GitHub Copilot: Performance Gains, Comprehension Trade-offs, and Behavioral Predictors in Brownfield Programming.
Under submission, 58th ACM Technical Symposium on Computer Science Education (SIGCSE) [pdf]

RESEARCH & TEACHING EXPERIENCE

Graduate Research Assistant, Oregon State University	Sep 2023 – Present
<i>Seal Lab — empirical studies of GenAI tools for code comprehension and CS education</i>	
<i>Designs and runs controlled experiments; mixed-methods analysis (qualitative + quantitative)</i>	
Graduate Teaching Assistant, Oregon State University	2024 – 2025
<i>CS 362: Software Engineering II</i>	

SKILLS

Research Methods: Mixed-methods empirical research, controlled experiments, semi-structured interviews, thematic analysis, survey design, behavioral log analysis, statistical analysis

Programming: Python, Java, C

Web Development: HTML, CSS, React, Express, Flask

Databases: MySQL, PostgreSQL, MongoDB

GenAI / AI Tools: Claude Code, OpenAI Codex, Google Gemini, GitHub Copilot

Tooling: Git / GitHub, LaTeX

Languages: English (fluent), Mandarin Chinese (native)